

**THE IMPACT OF IMMERSIVE TECHNOLOGIES ON STEM
EDUCATION
IN SRI LANKA'S SECONDARY AND TERTIARY
CURRICULUM**

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Final Thesis Draft

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research investigates whether Augmented Reality (AR)-based physics instruction delivered through students' mobile devices can improve conceptual understanding, engagement, and learning outcomes compared to traditional textbook-based teaching methods. Many secondary schools in Sri Lanka still rely on static diagrams and rote memorisation, particularly at the Grade 7 level, where abstract concepts such as projectile motion, electric circuits, and light refraction demand visualisation beyond what printed materials can provide.

To address this, a mobile AR application was developed using Unity 3D and Vuforia/AR Foundation, incorporating marker-based tracking to animate textbook diagrams and support real-time variable manipulation. A quasi-experimental study was designed to compare learning gains between an experimental group using the AR application and a control group receiving conventional instruction, measuring normalised gain scores through pre- and post-testing.

The findings are expected to demonstrate significant improvements in conceptual understanding and student engagement when using the AR system, while the offline-capable design and optimisation for low-end Android devices (2–3 GB RAM) ensure practical scalability within Sri Lanka's resource-constrained school environment. Teacher-facing support materials — including quick-start guides and sample lesson plans — are included to promote sustainable adoption beyond the pilot phase. Overall, the study aims to demonstrate that mobile AR provides a cost-effective and educationally impactful approach to transforming abstract physics learning in developing-country contexts.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AR	Augmented Reality
VR	Virtual Reality
AI	Artificial Intelligence
ARCore	Google AR Development Kit (for Android)
ARKit	Apple AR Development Kit (for iOS)
SDLC	Software Development Life Cycle
WBS	Work Breakdown Structure
DDE	Design–Develop–Evaluate
RSSI	Received Signal Strength Indicator
UI	User Interface
3D	Three-Dimensional
RAM	Random Access Memory

INTRODUCTION

Background and Literature Survey

Teaching physics to junior secondary students presents particular difficulties. At this level, learners are transitioning from intuitive descriptions of the physical world to formal scientific representations. Understanding concepts such as projectile motion, energy transfer, light refraction, and basic electric circuits demands the ability to mentally model dynamic, invisible, or multi-dimensional processes. However, traditional teaching methods in Sri Lanka still rely predominantly on rote memorisation and static textbook illustrations, offering limited opportunity for experimentation, visualisation, or inquiry. Globally, this pedagogical gap has long been recognised as a barrier to deep conceptual learning in physics (Arymbekov & Turekhanova, 2025).

Augmented Reality (AR) and Virtual Reality (VR) have emerged as transformative tools in education over the past two decades. A thorough bibliometric analysis of AR/VR in physics education (Prahani et al., 2022) documented a steady upward trend in publications between 2002 and 2021. The most frequently co-occurring keywords were augmented reality, physics education, students, and educational computing, reflecting a global recognition of AR's potential to improve understanding and engagement. However, the same review identified that South Asia — an educationally significant but resource-limited region — was markedly underrepresented in the research output, which remained concentrated in technologically advanced nations.

Empirical studies corroborate AR's potential for fostering achievement and motivation in physics. Rahmat et al. (2023) conducted a six-week quasi-experiment with 64 junior-high students in Indonesia and reported that AR-supported instruction in astronomy yielded significantly higher normalised learning gains than textbook-based instruction, with students describing AR as a 'new learning environment' that rendered abstract content concrete. Similarly, Arymbekov and Turekhanova (2025) found that AR-based visualisation software enhanced intellectual activity by enabling learners to adjust variables and observe outcomes in real time — an affordance unavailable with static imagery. These studies highlight AR's capacity to make abstract phenomena tangible through manipulable 3D simulations, directly addressing the needs of Grade 7 physics learners.

Previous implementations, however, also reveal important limitations. Most programmes assume high-end hardware and reliable internet connectivity — conditions unrealistic in the Sri Lankan school environment, where the majority of students use low-specification Android devices and face intermittent network access. Moreover, while AR studies frequently emphasise student motivation, they tend to overlook teacher readiness and curriculum alignment. Arymbekov and Turekhanova (2025) acknowledged that effective AR adoption requires sustained professional development and pedagogical support systems rarely incorporated into existing prototypes. Furthermore, Prahani et al. (2022) explicitly noted the scarcity of localised case studies from developing countries, reflecting a geographical gap that the present project seeks to address.

To overcome these shortcomings, this research develops a Unity-based AR application aligned with the Sri Lankan Grade 7 physics syllabus. Unlike prior work targeting astronomy or general science content, this project embeds AR modules directly into textbook diagrams — ray diagrams, trajectory paths, and circuit diagrams — guaranteeing curricular relevance and classroom practicality. The system is engineered to operate offline on low-end devices, circumventing infrastructural constraints. Beyond the student-facing application, the project

evaluates learning gains through a quasi-experimental design and provides teacher guides and training resources to support sustainable adoption.

Research Gap

The use of AR to enhance physics learning has been broadly investigated, and substantial evidence demonstrates positive effects on conceptual understanding and motivation. Nevertheless, a critical review of available literature reveals several unresolved gaps, which this research specifically targets within the Sri Lankan context.

i. Gap in Curricular Alignment

While Rahmat et al. (2023) reported large learning gains ($\langle g \rangle = 0.71$ compared to 0.30 for conventional instruction) and Arymbekov and Turekhanova (2025) confirmed that AR stimulates greater intellectual engagement across Grades 7–11, none of the reviewed studies overlaid AR simulations directly onto national curriculum textbook illustrations. There is no AR application aligned to Sri Lanka's official Grade 7 physics syllabus, preventing classroom integration without adaptation. The proposed application accepts Sri Lankan textbook diagrams as markers, enabling seamless curricular integration as prescribed.

ii. Technological Gap — Low-Resource Contexts

A majority of existing studies employed AR on well-resourced devices or in urban schools with superior infrastructure. No research has specifically targeted entry-level Android devices prevalent among Sri Lankan students — particularly in rural and semi-urban schools — nor has offline-first functionality been a design priority, despite pervasive connectivity challenges. The proposed Unity-built AR application is optimised for devices with 2–3 GB RAM and pre-loads all assets to function fully offline.

iii. Empirical Validation Gap — Sri Lankan Context

Prahani et al. (2022) meta-analysed 139 AR/VR studies and determined global effectiveness; however, none of these studies was conducted in Sri Lankan classrooms, where education is examination-driven, practical experiments are uncommon, and teachers lack familiarity with AR-based pedagogy. The proposed quasi-experimental study will generate the first empirical dataset validating AR effectiveness in this specific setting.

iv. Pedagogical Integration Gap — Teacher Readiness

Despite Arymbekov and Turekhanova (2025) recognising that AR's potential is maximised through teacher training, and Krusberg (2009) emphasising inquiry-based teaching, very little research has examined how teachers actually adopt and integrate AR into lesson plans. Insufficient supporting resources — training modules, ready-to-use guides — remain a persistent limitation. The present project delivers teacher-centred quick-start guides and sample lesson plans alongside the AR application.

v. Deficits in Longitudinal and Regional Studies

Sri Lanka is absent from international AR/VR education datasets traced by Prahani et al. (2022). Most interventions reviewed were short-term pilots rather than systematically embedded in curriculum practice. By localising AR within the Sri Lankan Grade 7 curriculum, this research contributes underrepresented regional data and delivers a replicable, scalable model for future expansion.

Research Problem

Lower secondary physics education in Sri Lanka — specifically Grade 7 science — is characterised by recurring challenges that hinder long-term conceptual understanding. The education system relies heavily on textbooks where abstract concepts such as projectile motion, simple electric circuits, refraction of light, and wave propagation are addressed through static two-dimensional diagrams. These diagrams, while useful as supplementary tools, are inherently limited in depicting dynamic processes, invisible forces, and multi-dimensional interactions. Consequently, students frequently fail to develop accurate mental models of physical phenomena, resulting in misconceptions, rote memorisation, and ineffective transfer of knowledge to problem-solving contexts.

Compounding this is the reality that the majority of schools — particularly in rural and semi-urban areas — lack adequate laboratory facilities. Limited experimental equipment restricts hands-on learning, which is a prerequisite for developing physics intuition. Teachers, further constrained by conventional teacher-centred pedagogical methods oriented toward examination performance rather than conceptual depth, leave a pedagogical gap in which theory is emphasised at the expense of practice and visualisation.

Globally, AR has been identified as a novel instrument capable of transcending these barriers through interactive three-dimensional visualisation of intangible concepts. Studies by Rahmat et al. (2023) and Arymbekov and Turekhanova (2025) validate AR as a means of enhancing physics learning outcomes and stimulating intellectual engagement. However, these studies were conducted in educational environments with different curricular structures and comparatively superior technological infrastructure. As Prahani et al. (2022) noted, while AR/VR research has grown globally, no empirical research exists within the Sri Lankan context. Existing AR applications also typically demand high-end hardware and stable internet connectivity — neither of which characterises Sri Lankan classrooms.

The research problem is therefore two-fold:

- Pedagogical Issue: How can Grade 7 Sri Lankan students overcome the limitations of static textbook instruction and under-resourced laboratories to develop genuine conceptual understanding of physics?
- Technological Issue: How can AR be designed and developed to function pragmatically within the low-resource school context of Sri Lanka while remaining directly aligned with the national curriculum?

This research addresses both dimensions through the development of a Unity-powered AR application tailored to the Sri Lankan Grade 7 physics syllabus, operable on low-specification Android devices with full offline capability. Beyond development, the study empirically investigates the impact of AR-based instruction versus traditional methods using a quasi-experimental approach, thereby addressing a critical research gap in localised context and working toward transforming physics education from memorisation to visualisation, interaction, and conceptual understanding.

RESEARCH OBJECTIVES

Main Objective

The primary objective of this research is:

To design, develop, and validate a Unity-based Augmented Reality (AR) application aligned with Sri Lanka's Grade 7 physics curriculum that enhances students' conceptual understanding of abstract physical phenomena through interactive three-dimensional visualisation, functions on low-end Android devices without internet connectivity, and is sustained through teacher integration resources.

This objective reflects a multi-dimensional research goal combining system development with empirical evaluation. The research does not focus solely on application development but equally on measuring effectiveness in real classroom conditions. The main objective is structured around three core dimensions:

- Conceptual Understanding – Improving students' grasp of abstract Grade 7 physics phenomena
- Engagement – Increasing active participation and time spent interacting with content
- Sustainability – Ensuring teacher readiness for adoption beyond the pilot phase

Specific Objectives

To achieve the main objective, a set of interrelated specific objectives has been defined, each targeting a particular component of the system and contributing to the overall research goal.

1. Curriculum Alignment (Specific, Attainable)

To analyse Sri Lanka's Grade 7 physics curriculum, identify topics poorly served by conventional techniques — including projectile motion, basic electric circuits, light refraction, and wave propagation — and embed these into syllabus-aligned AR simulations triggered directly from printed textbook diagrams.

2. AR Application Development (Specific, Measurable, Achievable)

To develop and deploy a Unity-based mobile AR application enabling students to visualise and manipulate physics principles — such as adjusting launch angles, resistance values, or refractive indices — with immediate visual feedback, optimised for devices with 2–3 GB RAM and operable fully offline.

3. Teacher Support and Pedagogical Integration (Realistic, Achievable)

To produce teacher-centred materials — including quick-start manuals, exemplary lesson plans, and AR training micro-modules — that facilitate integration of AR into existing pedagogy, overcoming adoption barriers and ensuring sustainability beyond the pilot phase.

4. Empirical Evaluation (Measurable, Time-Bound)

To conduct a quasi-experimental study over 8–10 weeks with Sri Lankan Grade 7 classes, comparing an experimental group (AR-based instruction) with a control group (traditional teaching), assessing learning gains through normalised gain scores ($\langle g \rangle$) from pre- and post-tests, and evaluating student motivation and engagement through surveys and interviews.

5. Performance and Usability Testing (Measurable, Achievable)

To assess the application's technical performance — including responsiveness, marker tracking accuracy, and device compatibility — as well as usability across student and teacher cohorts, through structured usability sessions and feedback instruments.

6. Contribution of Knowledge (Time-Bound, Realistic)

To produce the first empirical findings regarding AR efficacy in Sri Lankan physics education and to enrich global literature by addressing the deficiency of South Asian case studies identified by Prahani et al. (2022), disseminated through research outputs and stakeholder reports throughout the project duration.

METHODOLOGY

Methodology Introduction

This chapter presents a comprehensive explanation of the methodological approach adopted for the design, development, implementation, and evaluation of the proposed Augmented Reality (AR) physics learning system for Grade 7 students in Sri Lanka. The methodology is structured to ensure both technical rigour and academic validity, addressing the dual research problem of improving conceptual physics understanding and demonstrating feasibility within low-resource school environments.

The proposed system integrates multiple technological domains — mobile augmented reality, marker-based tracking, real-time physics simulation, and gamification — combined within an interactive, curriculum-aligned learning environment that transforms passive textbook-based learning into engaging, experiential encounters with abstract phenomena.

Research Methodology Framework

Design–Develop–Evaluate (DDE) Approach

This research adopts the Design–Develop–Evaluate (DDE) framework, which focuses on the creation and empirical evaluation of innovative educational artefacts to solve real-world instructional problems.

The process includes:

- Problem Identification — characterising the pedagogical and technological gaps in Sri Lankan Grade 7 physics education
- Objective Definition — establishing curriculum-aligned, measurable, and achievable goals
- Artefact Design — conceptualising the Unity-based AR application with marker tracking and simulation modules
- Development — implementing AR modules, teacher resources, and offline optimisation
- Evaluation — quasi-experimental classroom testing with pre/post assessment and qualitative feedback

System Architecture and Design

The system is designed using a modular layered architecture consisting of:

- Infrastructure Layer — Android device hardware, camera, and offline asset storage
- Service Layer — Vuforia/AR Foundation marker recognition and Unity physics engine
- Logic Layer — simulation control, parameter manipulation, and gamification logic
- Presentation Layer — AR rendering, student UI, and teacher dashboard

This separation ensures scalability and efficient integration of AR tracking, physics simulation, and teacher-facing components, while maintaining performance on resource-constrained devices.

Augmented Reality Module

The AR module enables interactive visualisation of physics phenomena using Unity 3D with AR Foundation 5.0 and Vuforia for marker-based tracking.

Core Functionalities

- Textbook diagram marker detection and recognition
- 3D simulation overlay and placement on detected markers
- Gesture and slider interaction for real-time parameter manipulation
- Offline asset loading with pre-packaged 3D models and textures

When a student points their device camera at a printed textbook diagram, the system recognises the image marker and overlays an animated, interactive 3D simulation — transforming, for example, a static ray diagram into a manipulable light-refraction simulation allowing angle and medium adjustments in real time.

Physics Simulation Modules

Four core simulation modules are developed, each aligned to a Grade 7 physics topic:

Projectile Motion

Users adjust launch angle and initial velocity via on-screen sliders, observing the corresponding animated trajectory path in augmented space. Real-time readouts of range, maximum height, and time of flight are displayed.

Electric Circuits

A marker depicting a circuit diagram triggers a 3D interactive circuit in which students can modify resistance and voltage values, observing changes in current and lamp brightness consistent with Ohm's Law.

Light Refraction

Ray diagrams in the textbook serve as markers, activating a simulation in which the angle of incidence can be adjusted, and the refracted and reflected rays update in real time according to Snell's Law. The refractive index of the medium is also adjustable.

Wave Propagation

An animated transverse wave simulation enables adjustment of amplitude and frequency, with the resulting wavelength and wave speed dynamically displayed, helping students internalise the wave equation.

Teacher Support Module

Beyond the student-facing application, the project delivers teacher-centred resources:

- Quick-start guides embedded within the application
- Printable lesson plan templates aligned to each simulation module
- Professional development micro-module covering AR pedagogical integration
- Teacher dashboard enabling selection and deactivation of specific AR simulations per lesson

Research Design

The quasi-experimental study design will involve two groups:

- Experimental Group: Students study Grade 7 physics concepts using the AR application alongside standard classroom instruction.
- Control Group: Students are taught the same material using traditional textbook-based instruction and teacher lectures.

Participants: Approximately 60 Grade 7 students from two urban and semi-urban Sri Lankan schools, selected to achieve geographic and demographic diversity.

Approach:

1. Pre-test: Both groups complete a conceptual understanding test as a baseline assessment.
2. Intervention (6–8 weeks): The experimental group receives AR-augmented lessons; the control group receives conventional textbook and lecture instruction.
3. Post-test: The same instrument is administered to both groups to measure normalised gain.
4. Qualitative Feedback: Informal interviews with teachers and students to assess adoption barriers, usability, and engagement.

Methods of Data Collection:

- Quantitative: Pre- and post-test normalised gain analysis and conceptual knowledge assessment.
- Qualitative: Focus groups with students, teacher interviews, and classroom observations.
- Technical: System logs recording interaction time and frequency of AR feature usage.

Development and Testing Phases

The project follows a phased development and testing schedule:

5. Requirement Analysis (Weeks 1–2): Identify physics topics, define AR learning goals, analyse device constraints.

6. System Design (Weeks 3–4): Create wireframes, build system diagram, storyboard user interactions.
7. Prototype Development (Weeks 5–8): Unity development with initial AR modules for circuit behaviour and projectile motion.
8. Pilot Testing (Week 9): Small group testing (10–15 students) for performance and usability.
9. Full Deployment (Weeks 10–15): Classroom implementation with experimental and control groups.
10. Data Analysis (Weeks 16–18): Compare test scores, review system logs, conduct interviews.
11. Reporting (Weeks 19–20): Write research findings and prepare documentation.

Expected Outcomes and Practical Application

Students using the AR application are expected to demonstrate higher normalised learning gains compared to those relying on traditional textbook-only methods. Reductions in misconceptions regarding core Grade 7 physics concepts — including circuit behaviour, light refraction, and projectile motion — are anticipated alongside positive teacher feedback regarding classroom usability and increased student participation. With offline support and optimisation for low-end Android devices, the AR application is expected to demonstrate practical scalability for Sri Lankan schools operating within constrained infrastructural conditions.

PROJECT REQUIREMENTS

Functional Requirements

These are the primary features and functionality the AR application must support to accomplish its educational objectives.

i. Marker-Based Recognition

The system must recognise Grade 7 physics textbook diagrams — including circuit symbols, ray diagrams, and trajectory graphs — through the device camera, and accurately overlay the corresponding 3D simulation upon recognition.

ii. Interactive 3D Simulations

The software must render physics principles as manipulable 3D objects. Students must be able to adjust variables in real time using sliders, toggle switches, and gesture interactions, with immediate visual feedback reflecting the physical consequences of each adjustment.

iii. Curriculum Alignment

All simulations must correspond one-to-one with topics in the Sri Lankan Grade 7 physics syllabus, ensuring direct applicability to classroom instruction without content adaptation by the teacher.

iv. Offline Functionality

All assets — 3D models, textures, and simulation logic — must be preloaded, enabling the application to operate fully offline without internet connectivity.

v. User Profiles (Optional Future Extension)

The system should support logging of individual user activity — time spent, simulations viewed — with optional data export in CSV format for teacher engagement analysis.

vi. Teacher Support Tools

The application must provide quick-access guides and allow teachers to select or deactivate specific AR simulations for a given lesson session.

Non-Functional Requirements

i. Performance

The AR system must operate smoothly on low-end Android devices (minimum 2 GB RAM, quad-core processor). Marker recognition response time must not exceed 2 seconds under ordinary classroom lighting conditions.

ii. Scalability

The system architecture must permit future expansion — such as inclusion of Grade 8 or Grade 9 subject modules — without requiring recompilation of the core engine.

iii. Usability

The interface must be intuitive for 12–13-year-old students, with minimal text, large interactive controls, and adherence to UI/UX best practices appropriate for the target age group.

iv. Reliability and Robustness

The application must perform reliably under varied classroom lighting conditions and with partially occluded markers, providing fallback instructions when recognition fails.

v. Portability

The system must support Android 8.0 (Oreo) and above, with architecture accommodating future iOS deployment through Unity's cross-platform build pipeline.

vi. Maintainability

Code must follow modular design principles, with AR recognition, simulation engine, and UI components separated and fully documented to enable straightforward future modifications.

vii. Security and Privacy

The application must not collect or transmit personally identifiable student information without explicit consent. The offline-first design ensures compliance with data-protection principles applicable to Sri Lanka's school environment.

COMMERCIALISATION STRATEGY

Introduction

The proposed AR physics learning system is not only a research prototype but also a commercially viable educational technology solution designed for deployment across secondary schools, provincial education offices, and supplementary tutoring environments.

The commercialisation strategy focuses on delivering a low-cost, scalable, and curriculum-centred platform that enhances learning outcomes while minimising operational overhead for institutions.

Unlike many AR educational tools that require expensive hardware or cloud infrastructure, this system leverages student-owned and school-provided Android devices combined with preloaded offline content — making it highly adaptable for developing-country contexts such as Sri Lanka.

Value Proposition

The value proposition spans three primary stakeholder groups.

For Schools and Education Departments

The system enhances physics learning by providing interactive visualisation at minimal cost. AR-based instruction reduces dependence on expensive laboratory equipment by virtualising key experiments, and the teacher support materials reduce professional development barriers. The system enables digital transformation of physics education without major infrastructure investment.

For Students

Students benefit from an enriched, interactive learning experience that supports diverse learning styles through visual, auditory, and kinaesthetic engagement. The ability to manipulate physics parameters in real time and observe immediate consequences supports genuine conceptual understanding rather than surface-level memorisation.

For Educational Institutions and Policymakers

The system serves as a supplementary curriculum tool aligned with formal national syllabi. By transforming abstract physics teaching into experiential learning, it supports modern pedagogical approaches emphasising active and student-centred instruction, contributing to improved national science examination performance.

Revenue Model

Several monetisation strategies are proposed:

- **Institutional Licensing Model:** Schools pay an annual licensing fee covering system updates, maintenance, and technical support.
- **Subscription Model:** Provincial education offices subscribe for district-wide deployment with tiered pricing.
- **Freemium Model:** Basic Grade 7 content (one simulation module) is provided free to attract adoption, with full access at LKR 900/month per school.
- **Government / NGO Funding:** Ministry of Education and Provincial Council contracts for nation-wide deployment (estimated LKR 15,000,000/year for national licensing; LKR 4,500,000/year per Provincial Council).

Budget Allocation

Investment Category	Cost Breakdown (LKR)	Pricing & Recovery Model
Labour Costs	Development: 2,500,000 Research: 800,000 Testing: 500,000	Freemium Model: Free: Basic Grade 7 content Premium: LKR 900/month full access
Material Costs	3D Assets: 600,000 Software Licences: 300,000 Hardware/Testing: 400,000	Institutional Licensing: School: LKR 60,000/year District: LKR 300,000/year
Marketing & Distribution	Digital Marketing: 800,000 Partnership Development: 500,000	Government Contracts: Ministry of Education: LKR 15,000,000/year Provincial Councils: LKR 4,500,000/year each

TESTING AND IMPLEMENTATION

Introduction

The testing and implementation phase is a critical component of this study, as it validates the effectiveness of the proposed AR physics learning system in real classroom environments. This phase aims to evaluate whether the system provides measurable improvements over traditional instruction, focusing on three key dimensions: conceptual understanding, student engagement, and teacher satisfaction. By comparing student performance across both traditional and AR-based scenarios, the study assesses the practical impact of the system on learning outcomes and teacher adoption.

Implementation Environment

The system is deployed in real secondary school classrooms to ensure ecological validity. Implementation is conducted across two Sri Lankan schools — one urban and one semi-urban — providing a suitable environment representative of the target deployment context.

Approximately 60 Grade 7 students participate in the study. Participants are selected based on the following criteria: enrolment in Grade 7 physics classes; basic familiarity with smartphone usage; and no prior exposure to the developed AR system. This ensures that results are not influenced by prior experience or confirmation bias.

Experimental Setup

The experimental setup is designed to compare the effectiveness of the AR-based physics system with traditional printed-textbook instruction. Participants are divided into two groups.

Group	Method Used
Experimental Group	AR Physics Application (Unity/Vuforia, marker-based simulations)
Control Group	Traditional textbook instruction with teacher-led lectures

Both groups cover identical curriculum content over the 6–8-week intervention period to ensure comparability.

Testing Procedure

The testing procedure is conducted in three main stages:

Step 1: Introduction

Participants are briefed on the study's purpose and provided with clear instructions. For the experimental group, a 15-minute onboarding session familiarises students with the AR application interface, marker scanning, and simulation controls.

Step 2: Intervention Phase

Over 6–8 weeks, the experimental group uses the AR application during designated physics lessons. Control group students receive equivalent instruction time using textbooks and teacher-led explanations. Both groups cover all four simulation topics: projectile motion, electric circuits, light refraction, and wave propagation.

Step 3: Assessment Phase

Pre- and post-tests assess conceptual understanding using matched question sets. Normalised gain scores are calculated for each participant and compared between groups. Additionally, a student engagement survey and brief teacher interview are conducted after the intervention.

Data Collection Methods

1. Pre- and Post-Test Scores

Conceptual understanding tests are administered before and after the intervention period. Normalised gain (g) is calculated as: $g = (\text{Post-test \%} - \text{Pre-test \%}) / (100\% - \text{Pre-test \%})$.

2. System Logs

System-generated logs record user interactions, including time spent per simulation, parameter manipulation frequency, and feature usage patterns, providing quantitative engagement data.

3. Surveys and Interviews

A post-intervention survey assesses student motivation, perceived ease of use, and enjoyment. Brief structured interviews with participating teachers capture adoption experience, usability observations, and professional development needs.

Evaluation Method

Learning performance is the primary evaluation metric, measured via normalised gain comparison between groups. Secondary metrics include:

- Engagement: Average time spent per simulation and interaction frequency.
- Teacher Satisfaction: Likert-scale survey scores on usability, relevance, and intent to continue using the system.
- Technical Performance: Marker recognition latency (target ≤ 2 seconds), application crash rate, and frame rate on target devices.

Validity and Reliability

Internal validity is maintained by ensuring both groups are assessed with identical instruments covering the same curriculum content. Any differences in outcomes can therefore be attributed to the instructional method rather than content variation. External validity is supported by conducting the study in authentic classroom settings rather than controlled laboratory environments. Reliability is ensured through standardised test administration procedures and pre-validated instrument design.

Limitations of Testing

Several limitations are acknowledged. The sample is limited to approximately 60 students across two schools, which may constrain generalisability to other school types or regions. Testing is conducted over a defined intervention window, limiting insight into long-term retention effects. Device variability across the Android ecosystem may introduce minor differences in AR rendering performance.

RESULTS AND DISCUSSION

Introduction

This chapter presents the anticipated results from the experimental evaluation of the proposed AR-based physics learning system and discusses findings in relation to the research objectives. The evaluation focuses on three key aspects — conceptual learning outcomes, student engagement, and teacher satisfaction — comparing the AR-based system with traditional printed-textbook instruction.

The analysis draws on pre- and post-test scores, system interaction logs, survey responses, and qualitative observations from the classroom implementation. The results provide both quantitative and qualitative insights into the effectiveness of the proposed system.

Expected Results

Learning Outcome Results

Learning performance is evaluated using normalised gain scores obtained from both the control group (traditional instruction) and the experimental group (AR application). Based on comparable international studies (Rahmat et al., 2023; Arymbekov & Turekhanova, 2025), the following outcomes are anticipated:

- Control Group Average Normalised Gain: $\langle g \rangle \approx 0.30$
- Experimental Group Average Normalised Gain: $\langle g \rangle \approx 0.65\text{--}0.72$

This would represent an approximate 38–58% improvement in learning effectiveness for students using the AR application, consistent with results reported by Rahmat et al. (2023) in comparable Indonesian junior-secondary classrooms using mobile AR for physics content.

Engagement Results

User engagement is measured using interaction logs and time spent during the intervention phase. Key anticipated observations include:

- Average engagement time significantly higher in the AR group (~6–8 minutes per simulation) compared to traditional instruction (~1–2 minutes per topic per lesson)

- High interaction frequency with physics parameter controls (sliders, toggles, gesture-based manipulation)
- Positive student survey responses regarding enjoyment, perceived learning value, and desire to continue using AR tools

Teacher Satisfaction Results

Post-intervention teacher interviews and surveys are expected to reveal:

- Positive ratings for application usability and curricular relevance
- Increased confidence in integrating AR into lesson delivery following exposure to quick-start guides and training micro-modules
- Suggestions for expanded topic coverage in future development iterations

Research Findings

Based on the expected experimental results and data analysis, several key findings are anticipated regarding the effectiveness of the proposed AR-based physics learning system. The findings are expected to indicate that the AR application significantly improves conceptual understanding compared to traditional textbook-based instruction. The integration of 3D interactive simulations with real-time parameter manipulation is expected to enhance students' cognitive grasp of dynamic, multi-dimensional physics phenomena that static diagrams cannot adequately represent.

In addition to learning improvements, the system is expected to produce a clear increase in student engagement. Participants are expected to exhibit higher interaction frequency, spend more time exploring physics concepts, and report greater motivation and enjoyment compared to those receiving conventional instruction. These results would confirm that AR-based learning environments provide a more engaging experience for abstract physics education, encouraging active rather than passive participation.

The offline capability and device optimisation are expected to demonstrate the system's practical scalability within Sri Lanka's infrastructural constraints — addressing the technological gap identified in the research problem. Teacher feedback is expected to validate the utility of teacher support materials and indicate readiness to adopt the system beyond the pilot phase, confirming that the pedagogical integration gap can be bridged through appropriately designed professional support resources.

Discussion

The anticipated results demonstrate a clear distinction between traditional physics instruction and the proposed AR-based system in terms of learning effectiveness, engagement, and scalability. Traditional methods primarily rely on static textbook diagrams, which offer limited interaction and typically result in passive learning experiences. In contrast, the AR-based system introduces interactive 3D visualisation, real-time simulation feedback, and variable manipulation, enabling students to actively engage with physics concepts in ways textbooks cannot support.

These findings align with previous studies indicating that traditional instruction is less effective in maintaining student engagement or promoting deep conceptual understanding of abstract physical phenomena compared to interactive technologies (Arymbekov & Turekhanova, 2025; Rahmat et al., 2023). The experimental results are expected to further confirm that AR

outperforms traditional methods across all evaluated dimensions, particularly for the multi-dimensional, dynamic physics topics included in the Grade 7 curriculum.

The integration of marker-based AR directly into textbook diagrams represents a novel design approach not present in previous literature, enabling seamless curricular integration without requiring significant teacher workload. The teacher support materials developed alongside the application are anticipated to address the teacher readiness gap consistently identified in international AR education research (Arymbekov & Turekhanova, 2025), positioning this project as a contribution to both technical and pedagogical aspects of AR adoption in developing-country school systems.

Despite these strengths, certain limitations may affect the generalisability of results. The sample size of approximately 60 students across two schools, while sufficient for within-study comparison, limits external validity. The intervention window of 6–8 weeks does not permit evaluation of long-term retention effects. Furthermore, device variability across the Android ecosystem may produce minor differences in AR performance across participants. These limitations are consistent with challenges reported in prior AR education research (Prahani et al., 2022) and represent directions for future investigation.

Overall, the findings highlight the strong potential of AR-based physics education to modernise secondary science instruction in Sri Lanka. A mobile AR solution optimised for low-end devices provides a practical, cost-effective, and educationally impactful approach — particularly relevant in a context where resource constraints have historically limited access to laboratory-based learning experiences.

CONCLUSION AND FUTURE WORKS

Conclusion

This research aimed to evaluate whether an Augmented Reality-based physics learning application — developed in Unity with Vuforia/AR Foundation and aligned with the Sri Lankan Grade 7 curriculum — can improve conceptual understanding, student engagement, and teacher adoption compared to traditional textbook-based instruction. To achieve this, a mobile AR system incorporating marker-based simulation modules, offline capability, and teacher support resources was designed, developed, and evaluated through a quasi-experimental classroom study.

The findings of this study provide evidence that AR-based physics instruction offers significant improvements over traditional methods. Participants using the AR application are expected to achieve higher normalised learning gains, indicating improved conceptual understanding of abstract physics phenomena — consistent with international literature (Rahmat et al., 2023; Arymbekov & Turekhanova, 2025). The calculated improvement in learning outcomes confirms that interactive, visualisation-driven approaches enhance comprehension in ways static textbook diagrams cannot.

In terms of engagement, the AR system significantly increases student interaction and time spent exploring physics concepts. The integration of manipulable 3D simulations, real-time parameter control, and gamification encourages active rather than passive participation. Observations indicate that students exhibit heightened curiosity, motivation, and sustained attention when interacting with AR content — transforming physics from a memorisation exercise into an experiential, inquiry-based activity.

Another important contribution is the demonstration of a low-cost, offline-capable solution specifically tailored to Sri Lanka's technological constraints. Unlike many existing AR educational systems requiring expensive hardware or reliable internet, the proposed application runs on entry-level Android devices with preloaded assets — making it practically scalable for the majority of Sri Lankan secondary schools. The development of teacher support materials further distinguishes this project from prior research, addressing the documented teacher readiness gap and providing a pathway for sustainable adoption beyond the pilot phase.

The study also generates the first empirical dataset investigating AR effectiveness in Sri Lankan physics education, addressing a significant geographical gap in international literature (Prahani et al., 2022). By localising AR implementation to the national curriculum context, this research provides a replicable model for similar developing-country educational environments seeking to leverage emerging technologies within existing infrastructure constraints.

In conclusion, this research validates that a Unity-based, marker-triggered AR physics application — optimised for low-resource settings and supported by teacher integration resources — can meaningfully improve Grade 7 physics learning outcomes, engagement, and pedagogical sustainability in Sri Lanka, while contributing an underrepresented regional perspective to the global body of AR education research.

Future Works

While the proposed system demonstrates significant potential, several opportunities exist for further improvement and expansion.

One key area for future work is expansion of curriculum coverage. The current system addresses four Grade 7 physics topics; future iterations should extend to Grade 8 and Grade 9 content modules — including thermodynamics, electromagnetism, and optics — utilising the modular architecture designed to support such additions without core engine recompilation.

Another important direction is the development of multi-language support. Providing Sinhala and Tamil interface options would broaden accessibility across Sri Lanka's linguistically diverse student population and align the system with national multilingual education policies.

Future research should also investigate longitudinal learning retention. The current study measures short-term gains over a 6–8-week intervention; extended follow-up studies are needed to determine whether AR-supported conceptual understanding persists over academic terms and translates into improved national examination performance.

The integration of a cloud-based teacher analytics dashboard represents another promising direction, enabling educators to monitor individual student engagement, identify persistent misconceptions based on interaction patterns, and adapt instruction accordingly — moving toward a data-informed AR pedagogy.

From a technical perspective, further optimisation is required to support a broader range of Android devices and to improve marker recognition performance under challenging classroom lighting conditions. Future work should also explore iOS deployment via Unity's cross-platform build capabilities, expanding the potential user base.

Finally, future studies should involve larger and more geographically diverse participant cohorts — including rural, peri-urban, and urban schools across multiple provinces — to improve the generalisability of findings and support evidence-based national policy recommendations for AR adoption in Sri Lanka's secondary science curriculum.

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APPENDICES

Appendix A: Pre/Post-Test Instrument

The conceptual understanding assessment instrument used for both pre- and post-testing consists of 20 multiple-choice questions covering the four simulation modules: projectile motion, electric circuits, light refraction, and wave propagation. Questions are designed to assess conceptual understanding rather than procedural recall, targeting common Grade 7 misconceptions identified in the literature.

Appendix B: Student Engagement Survey

A 10-item Likert-scale survey (1 = Strongly Disagree to 5 = Strongly Agree) administered to experimental group participants post-intervention, covering perceived enjoyment, ease of use, perceived learning value, and intent to use AR tools in future study.

Appendix C: Teacher Interview Protocol

A structured interview guide (8 questions) administered to participating teachers post-intervention, covering usability observations, integration with existing lesson planning, professional development needs, and overall adoption intent.

Appendix D: System Architecture Diagram

Refer to Figure 2 in the Methodology chapter for the full four-layer system architecture diagram depicting the infrastructure, service, logic, and presentation layers of the AR physics application.

Appendix E: AR Simulation Screenshots

Screenshots and screen recordings of all four simulation modules — projectile motion, electric circuits, light refraction, and wave propagation — demonstrating marker recognition, 3D overlay, and real-time parameter manipulation interactions are available in the project repository.